Thermal and Structural Feasibility of an
Uncooled SS316 Conical Convergent-Divergent Nozzle

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Portfolio Project 2026

Parameter	**Value**
Chamber pressure (Pc)	2 MPa
Chamber temperature (Tc)	800 K
Area ratio (ε = Ae/At)	4
Throat radius (Rt)	15 mm
Wall thickness (baseline)	4 mm
Material	SS316
Analysis type	One-way FSI (CFD → FEA)
Factor of safety (yield, Mesh 3)	42.7

Contents

1. Abstract

This study assesses whether uncooled SS316 stainless steel can survive the combined thermal and pressure loading at the throat of a small monopropellant convergent-divergent nozzle operating at $P_c = 2\ MPa$ and $T_c = 800\ K$. A one-way fluid-structure coupling maps CFD-computed wall pressure and temperature fields (ANSYS Fluent, 2D axisymmetric, k-ω SST) onto a structural model (ANSYS Mechanical) with temperature-dependent material properties, yielding throat von Mises stress and a factor of safety (FoS) against temperature-adjusted yield as the primary outputs. Isentropic hand calculations establish the operating point and validate the CFD to within 5% on throat Mach number, exit pressure, and mass flow. At the converged mesh, the throat carries 4.92 MPa, giving an FoS of 42.7 against yield; a wall-thickness sensitivity sweep (t = 3, 4, 5 mm) holds FoS > 33 throughout. The structure is therefore not yield-limited at this operating point. The binding constraint is instead creep: the wall sits at ~530 °C, above the SS316 creep threshold of ~410 °C. Uncooled SS316 is feasible for prototypes and testing, such as short-duration or ground-test firings, but sustained or flight-rated operation requires creep assessment.

2. Introduction and Project Scope

2.1 Engineering Question

Can an uncooled SS316 survive the thermal and pressure loading of a small monopropellant thrust nozzle with operating points of Pc = 2 MPa and Tc = 800 K?

2.2 Analysis Workflow

The analysis follows a six-phase sequential workflow.

Phase 1 Hand calculations: isentropic relations and the area-Mach equation (supersonic branch, selected by C-D geometry and choked-throat conditions) yield exit Mach, throat and exit static T/P, choked mass flow rate, and adiabatic wall temperature T_aw via turbulent recovery factor r ≈ 0.889.

Phase 2 Geometry: the inner aerodynamic profile is generated in SolidWorks with a 4 mm outward wall offset, preserving the area ratio for the CFD validation gate.

Phase 3 CFD: a 2D axisymmetric Fluent simulation resolves the wall static pressure and temperature distributions across the chamber, throat, and divergent section.

Phase 4 FEA: Fluent wall outputs are mapped onto the structural model via a one-way FSI handoff (External Data workflow, required by the 2D axisymmetric limitation), driving a coupled thermal-structural analysis in Mechanical with temperature-dependent SS316 properties. A wall thickness sensitivity study at t = 3, 4, and 5 mm follows in Section 7.

Phase 5 Convergence study: FEA results are checked across five QoIs (quantity of interests) for mesh independence while holding mesh variables constant except for the local size element at the throat.

Phase 6 Sensitivity study: the wall thickness is swept across t = 3, 4, and 5 mm at the converged mesh to test how throat FoS responds to thickness and to provide a fallback basis if the baseline proves to be marginal.

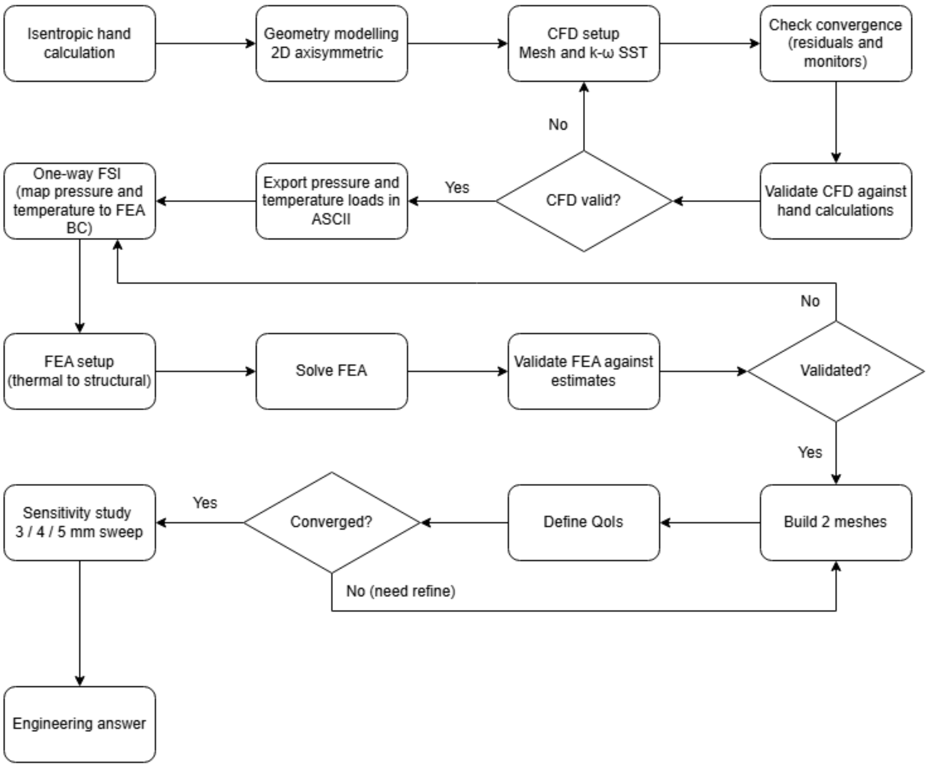


Figure 1. Analysis Workflow

2.3 Scope Boundaries

Analysis Coverage	**What This Analysis Does NOT Cover**
Steady-state thermal and structural response at one operating point	Transient startup/shutdown loading
Combined pressure + thermal gradient loading	Thermal fatigue / cyclic loading
Temperature-dependent SS316 properties	Creep at sustained elevated temperature
Von Mises stress, deformation, FoS, thermal gradient	Two-way FSI
3-mesh convergence study	Higher-fidelity Bartz convection BC follow-up
Wall thickness sensitivity (t = 3, 4, 5 mm)	Continuous thickness optimisation

One-way FSI (CFD → FEA)	Flight qualification (prototype/ground-test scope only)

3. Hand Calculations

3.1 Isentropic Flow Relations

The isentropic analysis assumes adiabatic, inviscid flow. With air as the working fluid ($\gamma = 1.4$, $R = 287$ J/kg·K) and the chamber as the stagnation reservoir, the operating point is defined by:

$$P_c = 2 \text{ MPa}$$

$$T_c = 800 \text{ K}$$

The area-Mach relation was solved numerically by scipy.optimize.fsolve to extract the supersonic exit Mach root. The initial guess of Mach 2 was declared to force the supersonic root.

Quantity	Value	Equations
Exit Mach number (Me)	2.940	$\frac{A_e}{A_t} = \frac{M_t}{M_e}\left[\frac{2+M_e^2(\gamma-1)}{2+M_t^2(\gamma-1)}\right]^{\frac{\gamma+1}{2(\gamma-1)}}$
Throat temperature (T*)	666.67 K	$T^* = T_c \cdot \frac{2}{\gamma+1}$
Throat pressure (P*)	1.057 MPa	$P^* = P_c\left(\frac{2}{\gamma+1}\right)^{\frac{\gamma}{\gamma-1}}$
Exit temperature (Te)	293.15 K	$\frac{T_c}{T_e} = \frac{2+M_e^2(\gamma-1)}{2}$
Exit pressure (Pe)	59,574 Pa	$\frac{P_c}{P_e} = \left(\frac{2+M_e^2(\gamma-1)}{2}\right)^{\frac{\gamma}{\gamma-1}}$
Mass flow rate (ṁ)	2.020 kg/s	$\dot{m} = \frac{A^* P_c}{T_c}\sqrt{\frac{\gamma}{R}}\left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{2(\gamma-1)}}$
Recovery factor (r)	0.889	$r = Pr^{1/3}$; turbulent BL (Pr = 0.703)
T_aw at throat	785.22 K	$T_{aw_t} = T^*\left(1 + r\left(\frac{\gamma-1}{2}\right)M^{*2}\right)$
T_aw at exit	743.83 K	$T_{aw_e} = T^e\left(1 + r\left(\frac{\gamma-1}{2}\right)M_e^2\right)$

Quantity	Value	Equations
Chamber radius (Rc)	42.4 mm	$\frac{A_c}{A_t} = \frac{M_t}{M_c}\left[\frac{2+M_c^{\ 2}(\gamma-1)}{2+M_t^{\ 2}(\gamma-1)}\right]^{\frac{\gamma+1}{2(\gamma-1)}}$

Although the chamber stagnation temperature is 800 K, the recovery factor $r \approx 0.889$ for a turbulent boundary layer indicates that a large portion of the kinetic energy is converted back to thermal energy near the wall. As a result, the adiabatic wall temperature reaches $T_{aw_{throat}} = 785.22\ K\ and\ T_{aw_{exit}} = 743.83\ K$, indicating that regardless of the gas expansion effect, these temperatures remain near the chamber temperature $T_c = 800\ K$. This suggests that the throat is the most thermally critical region due to the simultaneous presence of pressure, sonic Mach, and elevated wall temperature.

The chamber radius was determined from the contraction ratio $Ac/At = 8$, as per NASA SP-8120's recommended 5 to 15 range, which keeps the inlet Mach number below ~ 0.1, allowing the chamber to operate as the reservoir in CFD [7].

3.2 Nozzle Geometry

Among the nozzle geometry designs, the project is a conical convergent-divergent profile with a small throat fillet from the connecting-tangent lines. The conical profile uses straight-line walls through the converging and diverging sections, joined at the throat by a small fillet. It is chosen over the Rao bell contour for manufacturing simplicity, easier meshing, and consistency with 1D isentropic theory. However, there is a trade-off for the profile. The conical design encounters divergence loss due to a straight-angled contour spanning from the throat to the outlets, causing thrust inefficiencies. The effect does not influence structural analysis.

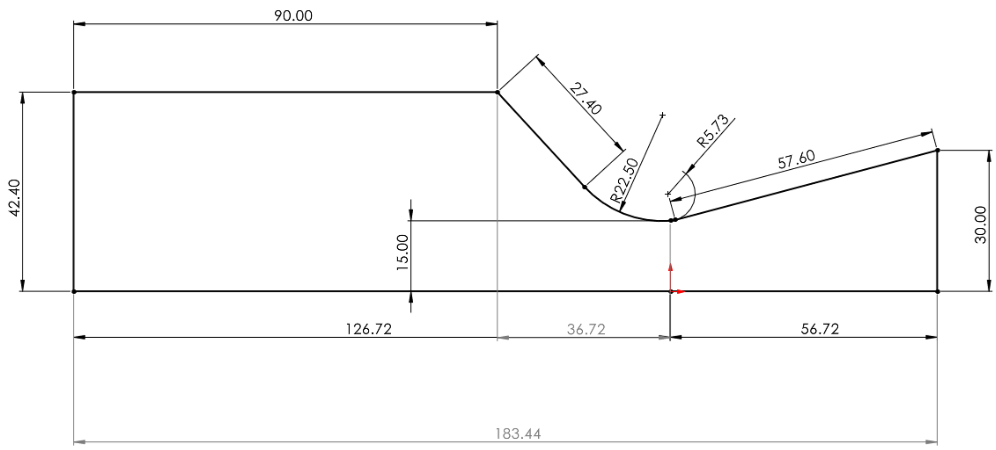


Figure 2. Nozzle cross-section with key dimensions

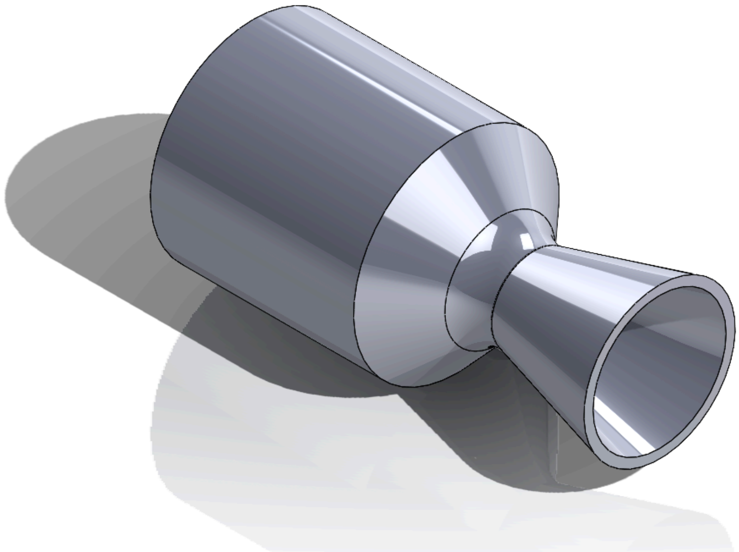


Figure 3. Nozzle isometric view

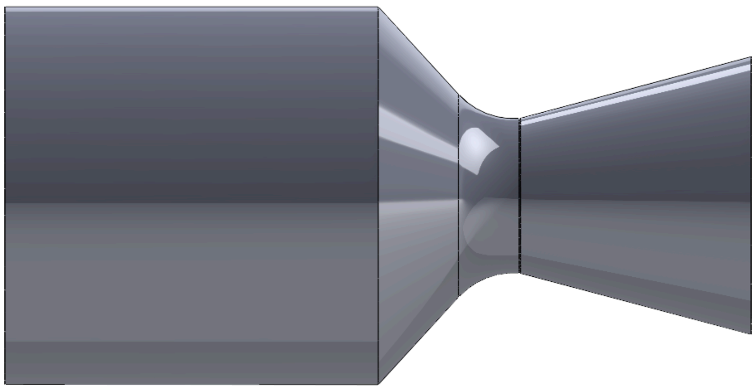


Figure 4. Nozzle right-side view

Geometry Feature	**Dimension**
Throat radius (Rt)	15.00 mm
Exit radius (Re)	30.00 mm
Chamber radius (Rc)	42.40 mm
Chamber length	90.00 mm
Convergent length (axial)	36.72 mm
Divergent length (axial)	56.72 mm
Total nozzle length	183.44 mm
Throat fillet radius	5.73 mm

Geometry Feature	Dimension
Divergent half-angle (~conical)	~14.8°
Wall thickness (baseline)	4.00 mm

4. Computational Fluid Dynamics (CFD)

4.1 Model Setup

Following the geometry and dimensioning, the CFD model for the controlled-volume domain was established. A 2D axisymmetric CFD model was used to simulate the nozzle flow in a density-based solver, which is a requirement for compressible supersonic flow conditions. The selected turbulence model used for the project is the k-ω SST, known for its capability in tackling solutions near wall.

The 2D axisymmetric simplification is valid here because the geometry is axisymmetric and the flow has no swirl. It is also necessary under the Student license cell limit, which a full 3D mesh would exceed.

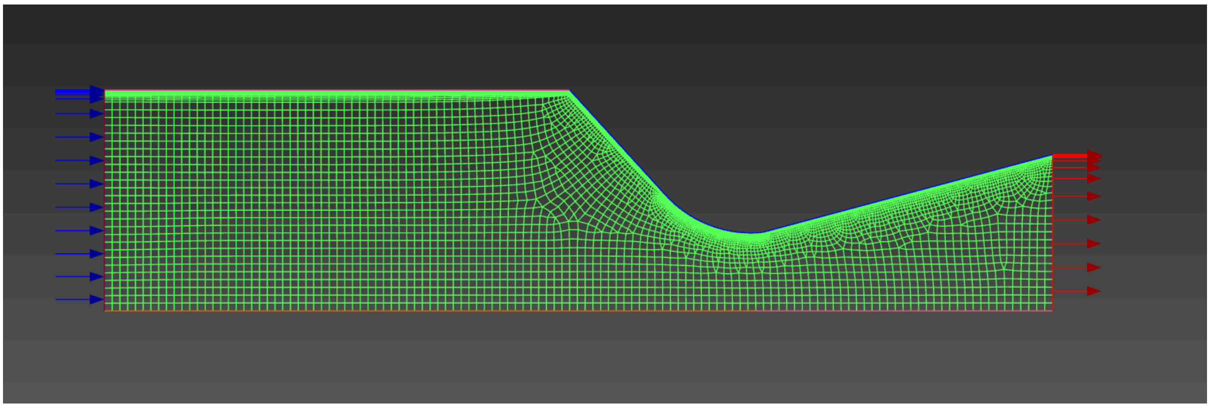


Figure 5. Boundary condition zones

Boundary	Condition Type	Value / Setting
Inlet	Pressure inlet	Pt = 2 MPa, Tt = 800 K
Outlet	Pressure outlet	Pe = 0.1 MPa
Inner wall	No-slip, adiabatic	$q_{wall} = 0$
Axis	Axisymmetric axis	—

4.2 Mesh

The mesh type is a 2D unstructured, quadrilateral-dominant hybrid mesh with a structured layer at the wall, generated in Fluent Meshing 2D workflow. The mesh setup is as follows:

Statistics	
Setting	**Values**
Total cells	15, 757
Quadrilateral fraction	97.7%
Min orthogonal quality	0.649
Max tangent skewness	0.878
Max aspect ratio	150
Min cell volume	9.15×10^{-11} m³
Max cell volume	5.56×10^{-7} m³
Sizing Controls	
Setting	Values
Global min	0.01 mm
Global max	1.5 mm
Throat curves	0.10 mm
Diverging wall	0.30 mm
Converging wall	0.50 mm
Boundary Layer	
Setting	Values
Offset method	Last-ratio
First layer height	0.01 mm
Number of layers	20
Growth rate	1.2
Max layer height	2 mm

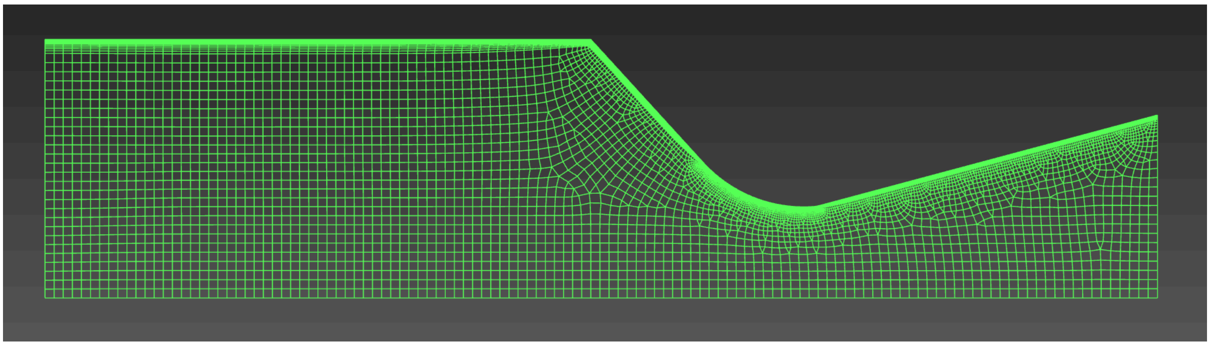


Figure 6. CFD Final Mesh

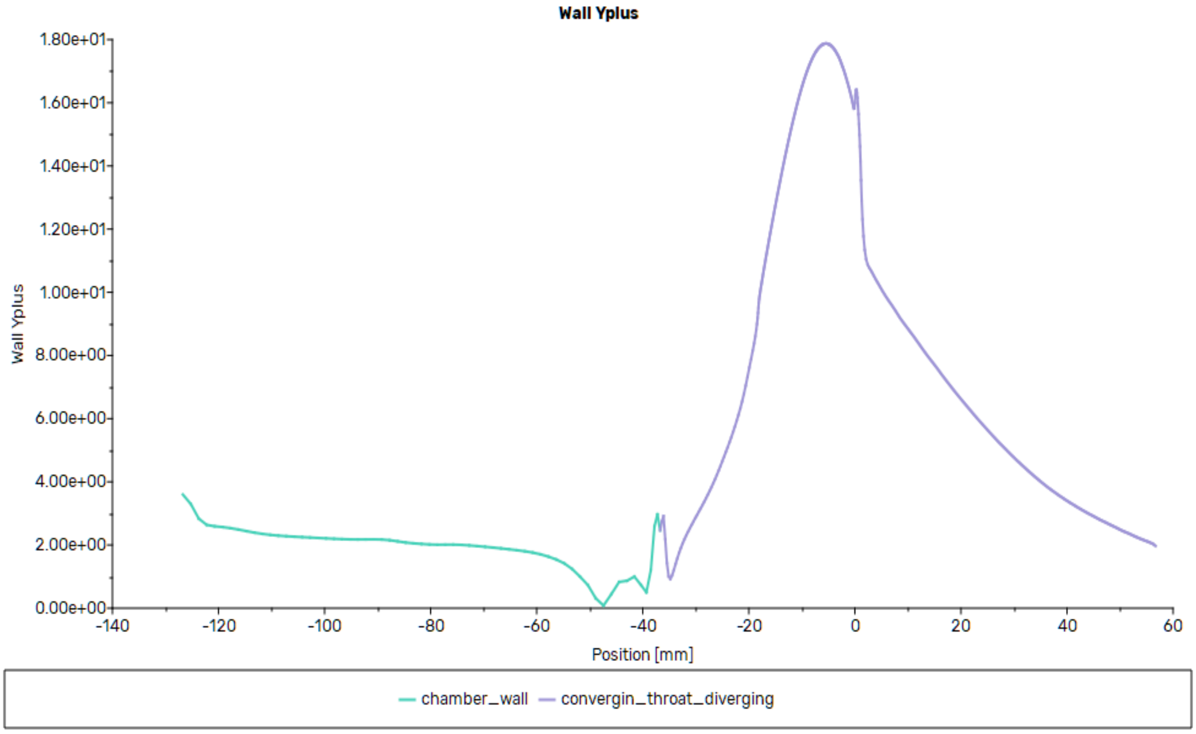


Figure 7. y+ distribution along the inner wall

y+ peaks at the throat (buffer zone) because of three effects: suppression of boundary layer growth by favourable pressure gradient (FPG), sonic section, and geometric curvature concentration. Because strong acceleration is developed from the chamber to the throat section, the boundary layer is repressed, suppressing the growth, and pushing the momentum closer to the wall. This is also accompanied by the velocity gradient and the curvature, resulting in a steeper velocity gradient and a higher shear stress at the wall.

4.3 Convergence

The CFD solver executed 11,000 iterations and demonstrated convergence for both flow and turbulence variables. The residuals plateaued at $\sim 10^{-3}$, a known behaviour for steady RANS at transonic conditions where the solver settles into a numerical limit cycle rather than driving residuals to round-off. This is confirmed by the converged k residual reaching its 10^{-5} target and, more importantly, by the mass-flow imbalance of 0.04%. Mass conservation, not residual magnitude, is therefore the governing convergence criterion.

Variable	Final residual
Continuity	$\sim 1 \times 10^{-3}$
x-velocity	$\sim 2 \times 10^{-3}$
y-velocity	$\sim 7 \times 10^{-4}$
Energy	$\sim 1 \times 10^{-3}$

k	~7×10^{-6}
ω	~2×10^{-4}

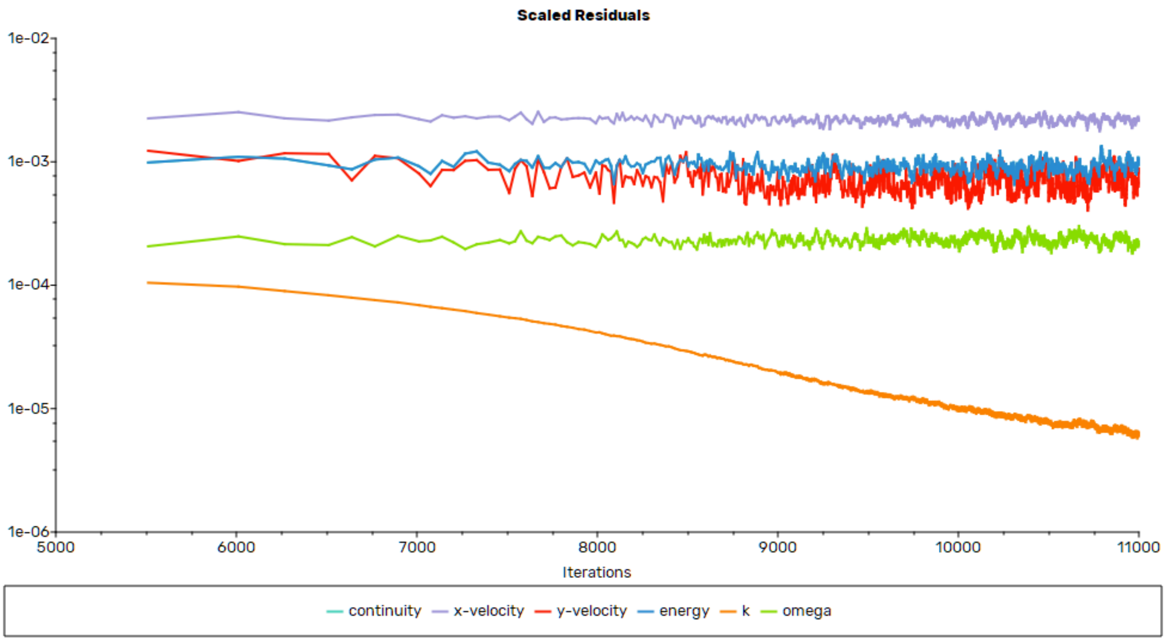


Figure 8. Residual history

4.4 Validation Gate

The CFD results are checked against the isentropic hand calculations to validate the solution. Four elements are checked, namely: the presence of sonic Mach at the throat, mass flow imbalance, exit Mach in area-weighted, and exit static pressure. The first confirmation of the validation is the presence of a sonic flow at the throat section, satisfying the supersonic flow downstream. Mass flow imbalance is the most critical indicator of a converged solution, since it directly tests mass conservation. Mach number and static pressure at the exit confirm supersonic flow and are validated against isentropic relations.

Check	CFD	Hand-calc	Deviation	Gate (≤5%)
Throat M = 1	Confirmed	1.0	—	Yes
Mass flow imbalance	0.04%	—	—	Yes
Exit Mach (area-weighted)	2.87	2.94	2.5%	Yes
Exit pressure (abs)	61,475 Pa	59,574 Pa	3.2%	Yes

4.5 Results

The contour shows the expected three-region structure: subsonic chamber, sonic line at the geometric throat, and supersonic expansion throughout the diverging section with no internal shocks. The Mach number peak of ~3.12 at the exit is the inviscid core value, not the bulk exit. The validation grade area-weighted exit Mach is 2.87, measured against the 1D isentropic theory of 2.94.

Figure 9. Mach number contour

Figure 10 shows static temperature throughout the domain. The chamber and the converging section sit near-isothermal at $T_c = 800\ K$, then the temperature drops sharply through the throat section and diverging section as thermal energy is converted to kinetic energy, reaching $\sim 277\ K$ at the exit ($T_e = 293.15\ K$). The diverging section also shows clear thermal stratification: a warmer sliver along the inner wall against a colder supersonic core, which is the visible signature of the adiabatic recovery layer where wall cells settle at T_aw. However, there is a flagged anomaly. A localised temperature overshoot appears on the gas-side wall just downstream of the throat (x ≈ 0.6–1.3 mm, r ≈ 15.1 mm), peaking at 808.2 K across a contiguous ~12-node band. This exceeds both the 800 K stagnation ceiling and the recovery-theory limit of T_aw = 785 K, so it is non-physical: static and adiabatic-wall temperatures cannot exceed stagnation for r < 1. The cause is near-wall energy-equation discretisation error in the buffer-layer cells, the same region where y+ peaks (L1).

Figure 10. Static temperature contour

Figure 11 shows the static pressure contour through the domain. A gauge pressure of ~1.90 MPa sits in the chamber (≈ 2.0 MPa absolute at the 100 kPa operating reference). A smooth pressure decay is also visible through the throat, with a minimum of -54.5 kPa. The -54.5 kPa is an anomaly, which is read as an isolated single-cell BC artefact. -54.5 kPa

Figure 11. Static pressure contour

Across all three fields, the solution reproduces the expected supersonic nozzle structure with no internal shocks, and the wall pressure and temperature distributions provide the validated boundary-condition inputs for the structural analysis in Section 5.

5. Finite Element Analysis (FEA)

5.1 Model Architecture

The FEA model uses a one-way fluid structure (FSI) chain. Wall static temperature and wall static pressure distributions are exported from the converged CFD solution as ASCII files and mapped onto the structural mesh through two External Data systems in ANSYS Workbench. The pressure file drives a Static Structural system (H); the temperature file drives a Steady-State Thermal system (G), whose nodal solution is linked into system H as a body-temperature load.

System	Function
F = Geometry	2D axisymmetric surface body (SpaceClaim)
G = Steady-State Thermal	Inner-wall temperature from CFD; outer-wall natural convection
H = Static Structural	Inner-wall pressure from CFD; body temperature from G; displacement constraint
I = External Data (Temperature)	Static temperature mapping; linked to G
J = External Data (Pressure	Static pressure mapping; linked to H

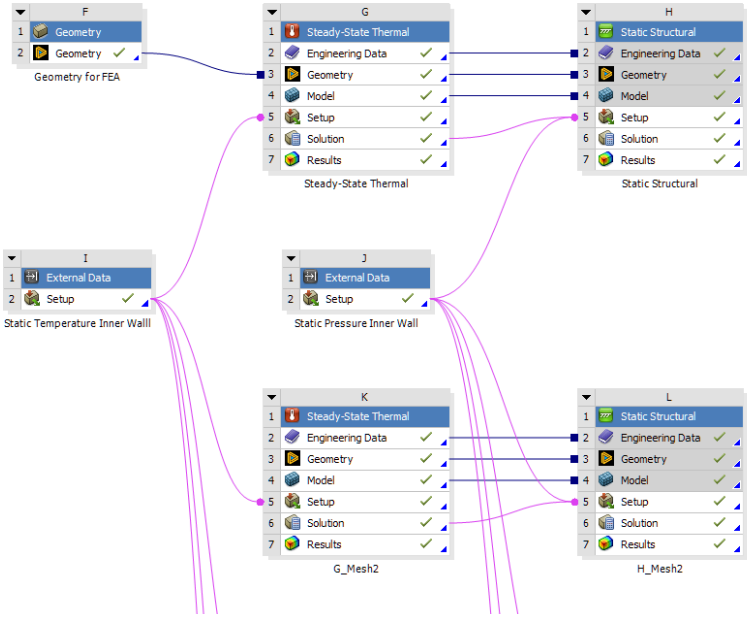


Figure 12.1.ANSYS Workbench project schematic part 1

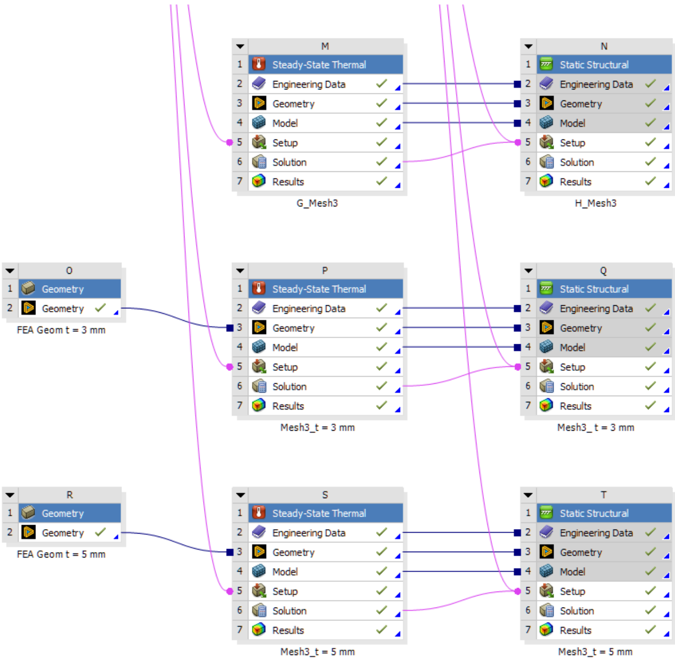


Figure 12.2. ANSYS Workbench project schematic part 2

Thermal boundary conditions of System G: the inner wall is held at the CFD adiabatic-wall temperature (T_aw, 768 K - 808 K). The outer wall uses natural convection with $h = 1 \cdot 10^{-5} W/mm^2 \cdot K$ (equivalently 10 W/m²·K) and $T_\infty = 300\ K$.

Structural boundary conditions H: the inner wall carries the imported pressure distribution - 0.04 MPa - 1.90 MPa. The body temperature field is inherited from the G solution. A single vertex at the inlet outer-wall corner is constrained in Y (axial) with X free (radial), removing the rigid-body axial mode without restricting radial thermal expansion. The constraint introduces a stress singularity at the inlet that is acknowledged and quantified.

Directly imposing T_aw: the adiabatic wall temperature is applied as a fixed boundary condition on the inner wall rather than coupling a heat-flux load with wall-side conduction to the gas.

5.2 Material Properties

The SS316 material properties were entered into ANSYS Engineering Data and are set up for temperature dependence with varying temperatures of 20 °C, 200 °C, 400 °C, and 600 °C.

The SSINA Table 1 / Nickel Institute 9004 source tabulates typical short-time yield strength at 200 °F intervals (Room, 400, 600, 800, 1000, 1200 °F). The 20/200/400/600 °C values in the table below are linearly interpolated onto even-Celsius brackets from these source rows.

Property	20 °C	200 °C	400 °C	600 °C	Units
Young's Modulus, E	200	186	172	152	GPa
Yield Strength, σ_y	310	263	238	195	MPa
Ultimate Tensile Strength, σ_u	621	553	496	399	MPa
Thermal Conductivity, k	14	16	18.6	21.5	$W/m \cdot K$
Thermal Expansion, α	16	16.5	17.5	18.5	$\times 10^{-6}/K$
Density, ρ	7950	7880	7801	7713	kg/m^3
Poisson's Ratio, ν	0.30	0.30	0.30	0.30	-

5.3 Mesh

The structural mesh is a 2D axisymmetric, quadrilateral-dominant mesh with quadratic (Tri6 + Quad8) elements, generated in ANSYS Mechanical.

Statistics	
Setting	**Values**
Sheet Body Method	Quadrilateral Dominant

Mesh sizing	Medium
Body sizing	1.5 mm
Wall edge sizing	0.5 mm
Throat curve sizing	0.3 mm
Thickness division	4
Nodes	8,196
Elements	2,489
Combined	10,685
Min element quality	0.674
Avg element quality	0.933
Std deviation	0.0432
Element types	Tri6 + Quad8

5.4 Boundary Conditions

The structural and thermal boundary conditions are listed below. Each load is imported from the converged CFD solution via the External Data workflow; the kinematic constraint and the outer-wall convection are applied directly in Mechanical.

Boundary	**Condition**	**Value**
Inner wall	Imported temperature, T_{aw}	$767.85\ K - 808.17\ K$
Inner wall	Imported pressure, pressure load	$-\ 0.039\ MPa - 1.898\ MPa$
Outer wall	Convection	$h = 10\ W/m^2 \cdot K; T_\infty = 300\ K$
Inlet vertex	X = free; Y = 0 displacement	Symmetry constraint
Reference temperature	20 °C across all systems	Engineering Data, body, environment

The imported boundary condition set was corrected across three (3) document revisions (v1, v2, and v3).

The outer-wall convection coefficient was entered in Mechanical as 10 in the active unit system $W/m^2 \cdot K$, and therefore six orders of magnitude too high, was corrected to $1 \cdot 10^{-5} W/mm^2 \cdot K$. The imported pressure field, mapped with its axial coordinates reversed so that the chamber and exit values were swapped, was corrected with a multiple of -1 on the Y-coordinate column, and the reference temperature, which was inconsistent between Engineering Data and Mechanical data default, was set to 20 °C across all systems. In v3, the temperature load was re-linked to the current CFD export after it was found to be reading an older file; this changed the imported temperature by less than 1 K and the throat factor of safety by about 3%.

The thermal load is applied as an imposed wall temperature equal to the CFD adiabatic recovery temperature. This follows a conservative approach because it holds the inner wall at the hottest temperature it could reach (with no heat conducted away) instead of solving the gas-to-wall heat transfer. In the sensitivity study, this is essential. The wall is held at T_{aw} regardless of the variation in thickness, and changing the wall thickness affects the structural margin more than it affects the temperature.

The kinematic constraint is placed at the outer-wall vertex of the inlet face. In a 2D axisymmetric model, the only rigid motion remaining to remove is sliding along the axis. A single Y = 0 constraint with X = free is enough to hold the model in place without stopping the wall from expanding outwards as it heats. The inlet was chosen so that the resulting stress spike of this constraint sits as far from the throat as possible. The spike is non-physical, shown in Phase 5 and demonstrates the global peak stress as the mesh is refined while the throat stress remains steady. Consequently, the FoS is taken from the throat, not the global maximum.

5.5 Results

Quantity	Value	Notes
Throat von Mises stress (probe)	5.03 MPa	Mesh 1 (baseline for convergence study)
Global max von Mises (singularity)	267.06 MPa	Inlet vertex constraint (non-physical, excluded from FoS)
Max total deformation	1.71 mm	Exit end (free); thermal expansion dominated
Through-wall ΔT at throat	2.12 K	small ΔT; outer-wall film resistance dominates the heat path
Throat temperature	804.38 K (531.23 °C)	Effectively at T_aw
Max thermal strain (X-axis)	9.36×10^{-3}	At throat region

The throat stress is about 5 MPa at this Mesh 1 baseline. This stress is essential, which is judged against the material SS316 yield strength at 531.23 °C. This results in a large safety margin and indicates that the wall is significantly far from yielding.

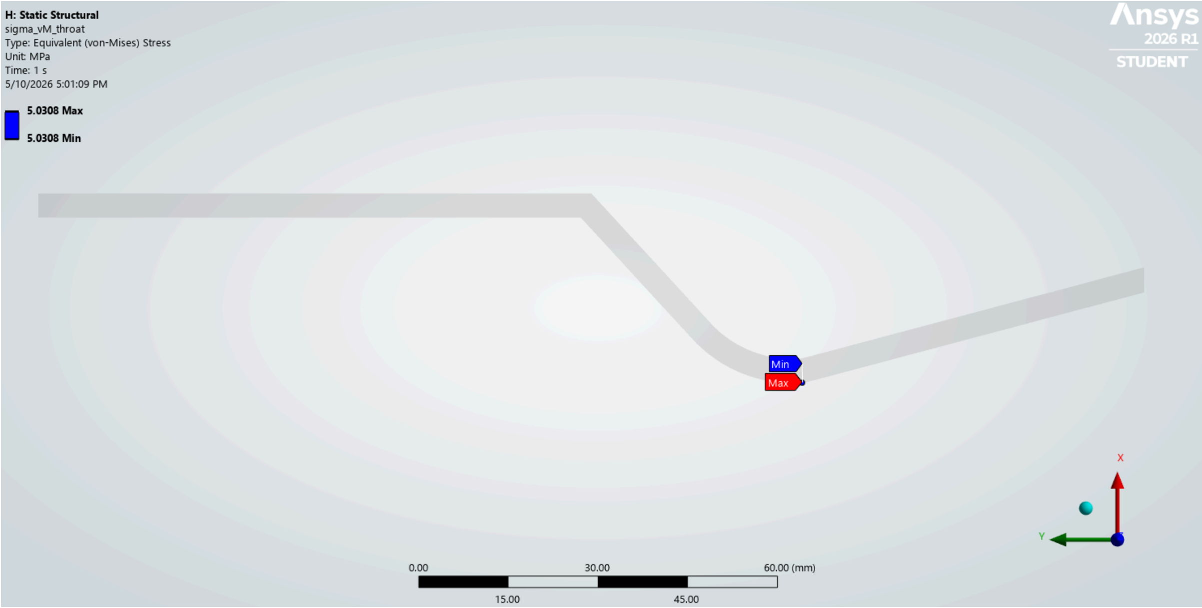


Figure 13. Von Mises stress probe at the throat

The global maximum of 267.06 MPa is a numerical artefact at the inlet constraint point, not a real stress. The continuous increase for subsequent mesh refinement is a concrete sign of a singularity. Hence, it is excluded from the FoS calculation.

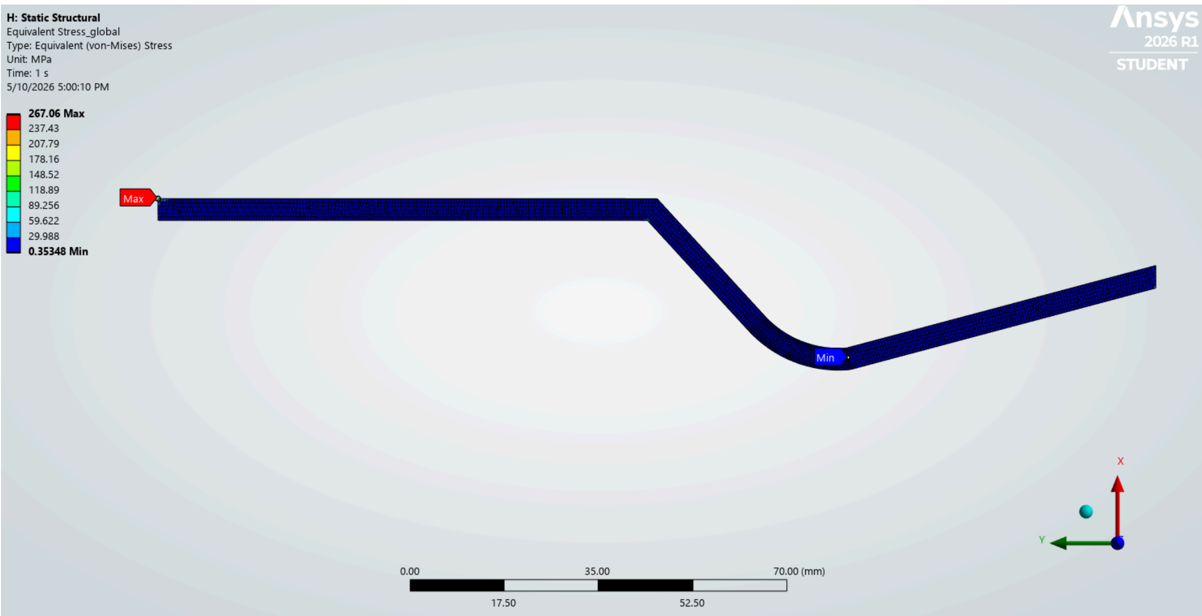


Figure 14. Von Mises global stress

The wall expands about 1.7 mm at the free exit end, driven by thermal expansion along the nozzle length. This affects how the nozzle mounts and seals, but it is not a structural failure.

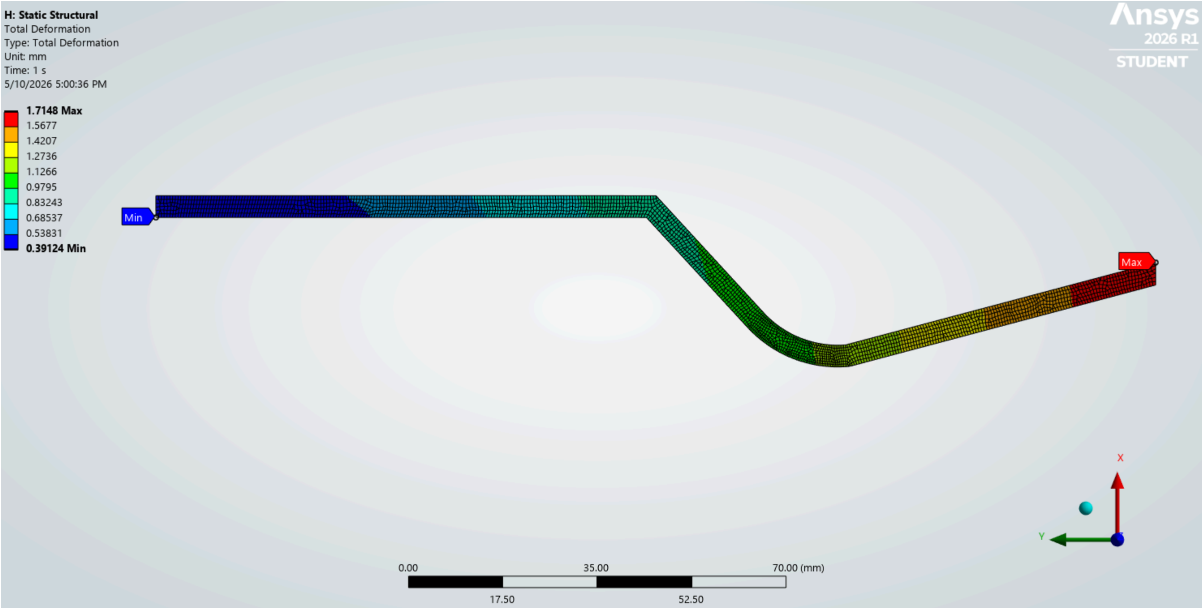


Figure 15. Total deformation contour

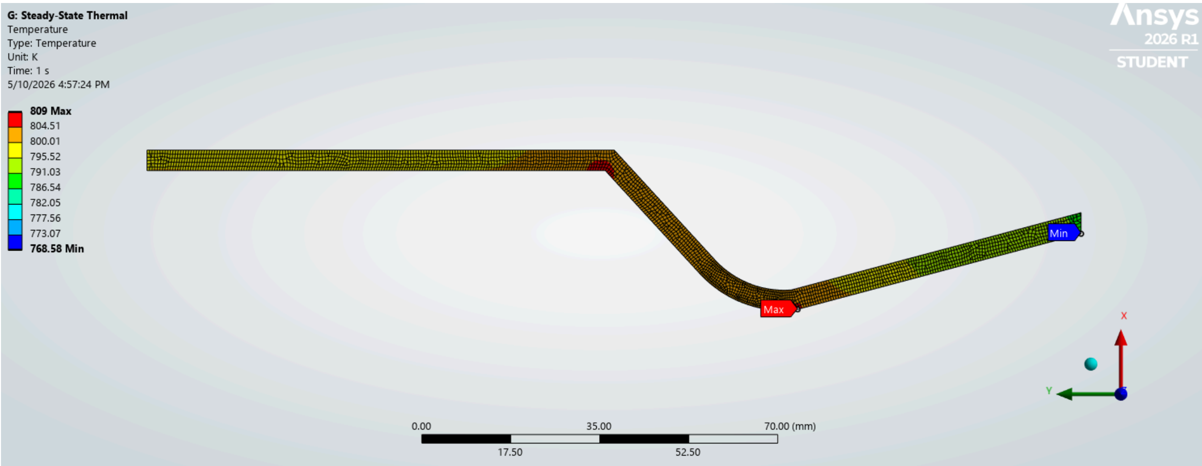


Figure 16. Steady-state temperature distribution

The through-wall temperature drop at the throat is only 2.12. Because the outer wall sheds heat slowly ($h = 10\ W/m^2 \cdot K$), the through-wall heat flux is small and only a ~2 K gradient develops; the wall sits almost uniformly at the gas recovery temperature.

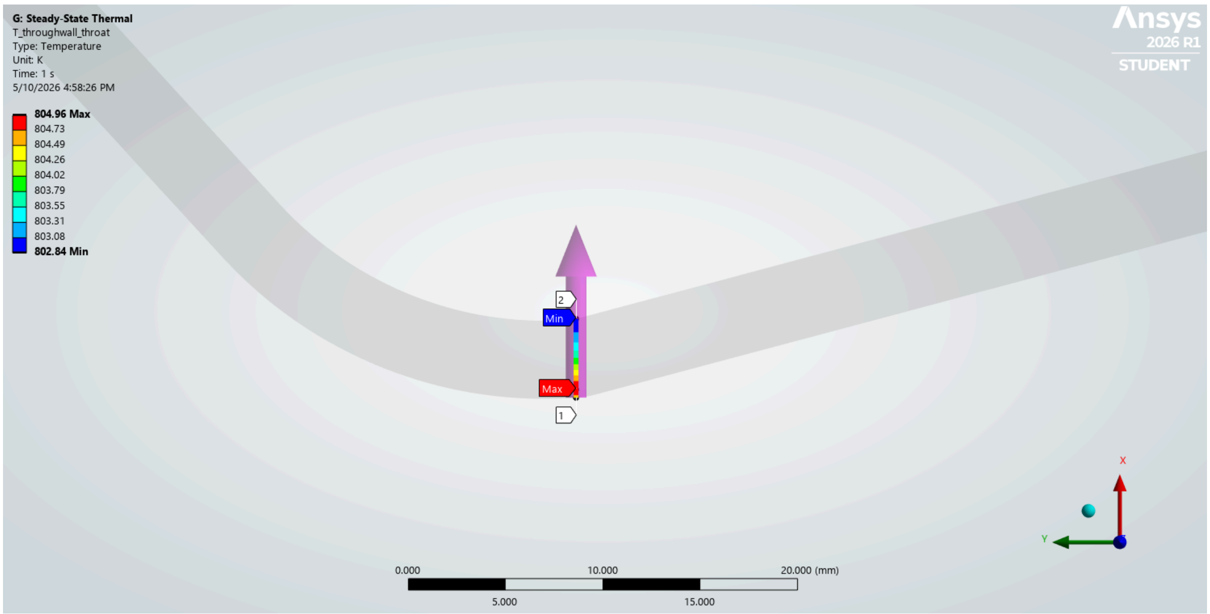


Figure 17. Through-wall ΔT at throat

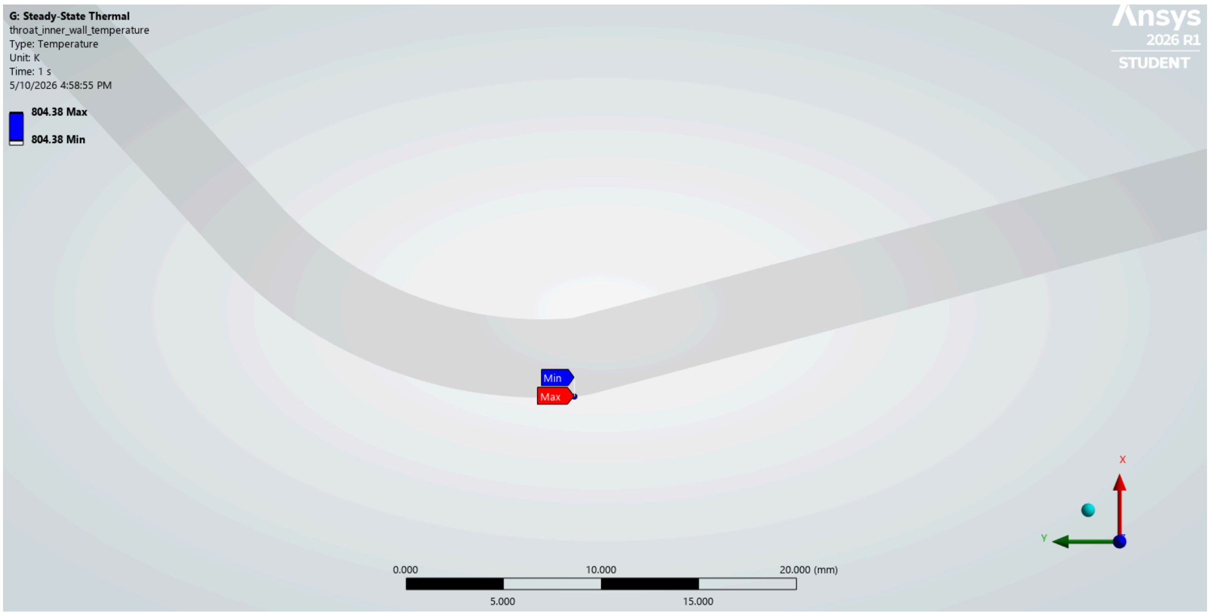


Figure 18. Temperature at the throat

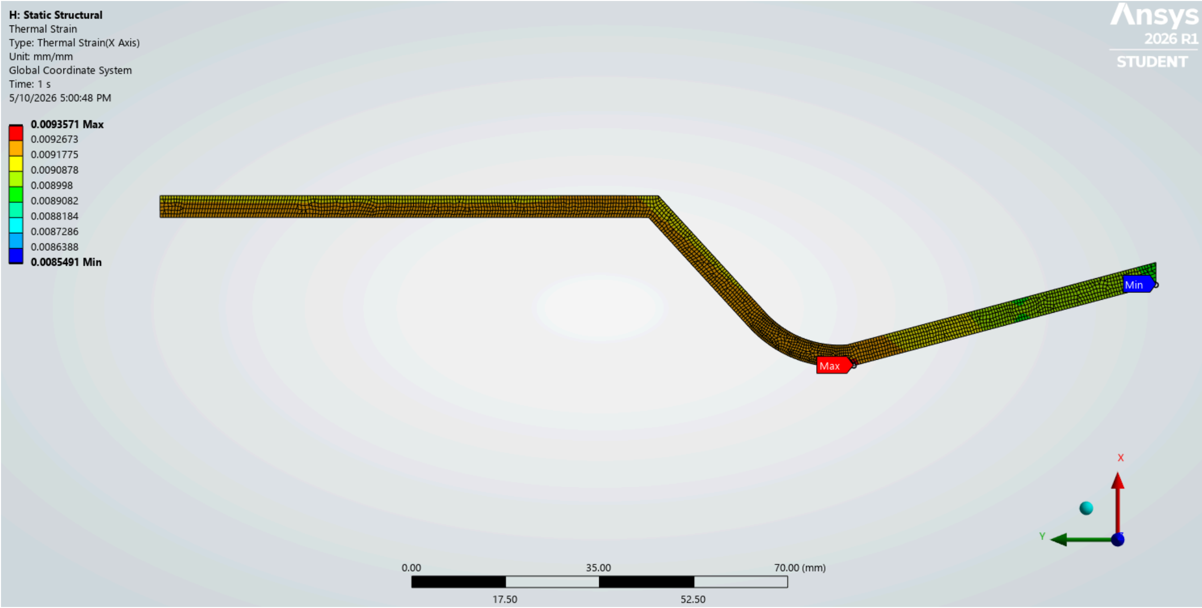


Figure 19. Thermal strain (X-axis) contour

5.6 Pre-Solve Validation Estimates

Before each solve, simple analytical estimates are computed and compared against the FEA results. These order-of-magnitude checks confirm each result is the right size and catch boundary conditions entered with the wrong units or sign before solving. This gate identified the convection units error (six orders of magnitude off) and the reversed pressure mapping from CFD pressure field imports.

Check	Analytical	FEA	Deviation
Through-wall ΔT (conduction-predicted)	~1–2 K	2.12 K	within range
Free axial expansion ($\Delta L = \alpha \cdot \Delta T \cdot L$)	1.60 mm	1.7148 mm	6.6%
Pressure hoop stress ($\sigma_\theta = P^* r/t$) P*·r/t at throat)	3.96 MPa	5.03 MPa	thermal-bending contribution
Imported Load Min/Max (v3 freshness gate)	T 768–809 K; P −0.04 to 1.90 MPa	matches	

The axial expansion uses the mean thermal expansion coefficient between the 20 °C reference and the 531.23 °C with $\alpha = 17.1 \times 10^{-6}/K$ and the actual nozzle length of 183.44 mm. The 6.6% is expected for a straight-bar estimate compared against the curved FEA wall.

The throat hoop stress uses the throat wall pressure of 1.06 MPa from the isentropic hand calculation. The hoop term alone under-predicts the axial end-cap stress and differential thermal expansion. The

~1 MPa gap is therefore expected and confirms the stress is the right size, not that hoop equals the full result.

The imported Load Min/Max is a data-provenance check, and it confirms the mapped load still reads the current CFD file.

6. Mesh Convergence Study

6.1 Study Design

This study uses a three-mesh trend approach to confirm that the headline results are mesh-independent and not artefacts of discretisation. The principle is to hold everything constant except for the variable being tested: local element size. Therefore, any change in the results can solely be attributed to mesh resolution.

Three meshes were generated using the same meshing method (quadrilateral-dominant), the same global element sizing (medium), and the same boundary conditions. The local element size is the primary variable of change along with the regions of interest between refined meshes, by a fixed linear refinement ratio. The three meshes produced combined node-plus-element counts of Mesh 1, Mesh 2, and Mesh 3 (10,685 / 16,527 / 21,565).

Five quantities of interest (QoIs) were tracked across all three meshes:

1. von Mises stress at the throat
2. Temperature at the throat wall
3. Through-wall temperature difference
4. Deformation at the free-exit end
5. Global von Mises stress

The von Mises stress at the throat is the FoS-critical metric on which the entire feasibility is dependent. The temperature at the throat wall is the primary thermal accuracy check that supports the scrutiny for the temperature-dependent yield strength. The through-wall temperature difference at the throat drives the thermal-gradient stress and is used to track and confirm the thermal field if it's resolved. The deformation at the free-exit end is used to check the magnitude of the structural response. Finally, the global von Mises stress is included as a positive control that sits at the outer-wall inlet face constraint singularity and is expected to keep rising as the mesh refines.

A QoI is taken as converged when the change between successive meshes falls below 5%. The four engineering QoIs are expected to conform to the gate.

6.2 Results

Four of the five QoIs converge to within 1% by the final mesh step, comfortably inside the 5% gate. The throat von Mises stress settles at 4.92 MPa with a final-step change of +0.92%. The throat temperature, through-wall temperature difference, and exit deformation are all mesh-independent within solver noise.

The global maximum von Mises stress behaves differently as it rises with every refinement, from 267 MPa to 353 MPa to 431 MPa, and shows no sign of settling. This is an expected behaviour of a constraint singularity located at the inlet vertex, where Y = 0 displacement boundary condition is applied. At a singularity, the true stress is mathematically unbounded, so a finer mesh simply resolves a sharper, higher peak.

This divergence is not a failure of the study, but rather a positive confirmation on two counts: first, it shows that the singularity is numerical rather than physical; a real stress concentration would converge to a finite value, whereas this one diverges precisely because it is an artefact of an idealised point constraint. Second, it demonstrates that the mesh is correctly resolving stress gradients, and the solver is responding to the refinement, which satisfies the prediction of convergence theory. Achieving four converged QoIs gives confidence that the results are not coincidentally flat. The global maximum is therefore reported but excluded from any engineering conclusion, and the throat probe is the engineering result.

QoI	Mesh 1	Mesh 2	Mesh 3	M1→M2	M2→M3
$\sigma_{VM_{throat}}$ (MPa)	5.0308	4.8725	4.9171	−3.15%	+0.92%
T_{throat} (K)	804.38	803.59	803.59	−0.10%	0%
ΔT_{throat} (K)	2.12	2.10	2.10	−0.94%	0%
δ_{exit} (mm)	1.7148	1.7149	1.7150	+0.006%	+0.006%
$\sigma_{VM_{global}}$ (MPa)	267.06	352.62	430.58	+32%	+22% Singularity

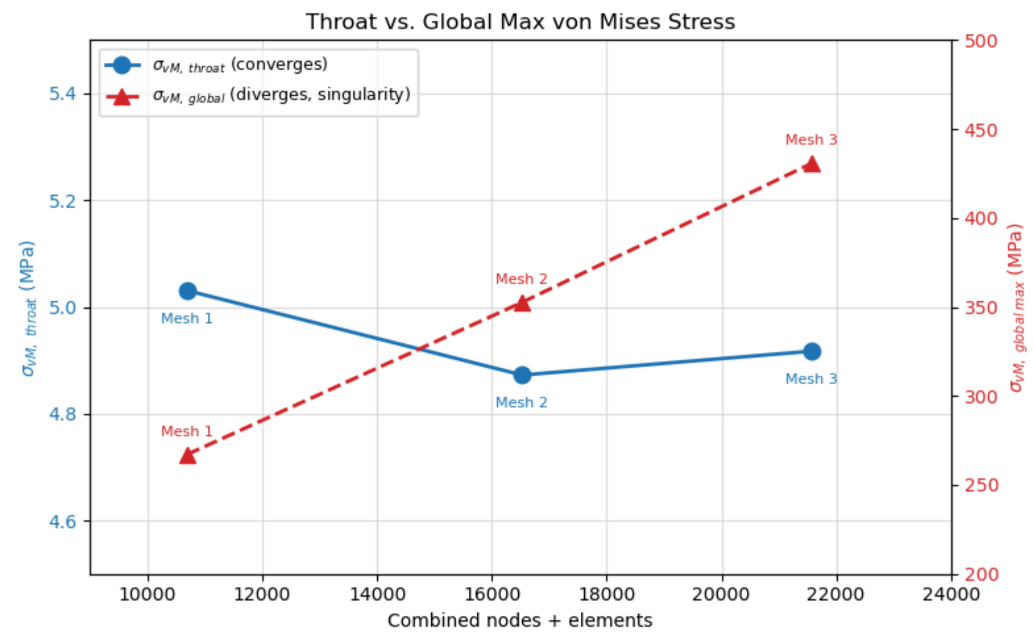


Figure 20. Convergence plot of von Mises stress (throat vs global)

6.3 Factor of Safety at Converged Mesh

The factor of safety is evaluated at the converged Mesh 3 result. Because the SS316 yield strength is temperature-dependent, the allowable stress must be read at the throat wall temperature.

The converged throat temperature is first converted from Kelvin to Celsius:

$$T_{throat,°C} = 803.59 - 273.15 = 530.44\ °C$$

The SSINA Table gives the SS316 yield strength at two bracketing temperatures: $\sigma_y = 238\ MPa$ at 400 °C and $\sigma_y = 195\ MPa$ at 600 °C [10]. Linear interpolation to the throat of 530.44 °C gives:

$$\sigma_y(530.44\ °C) = 238 + (195 - 238) \cdot \frac{530.44-400}{600-400}$$

$$\sigma_y(530.44\ °C) = 238 - 43 \cdot 0.6522$$

$$\sigma_y(530.44\ °C) = 209.95\ MPa$$

With the temperature-adjusted yield strength and the converged throat von Mises stress $\sigma_y = 4.9171\ MPa$ at Mesh 3:

$$FoS = \frac{\sigma_y}{\sigma_{vM_{throat}}} = \frac{209.95\ MPa}{4.9171\ MPa}$$

$$FoS = 42.7$$

7. Wall Thickness Sensitivity Study

7.1 Study Design

The wall thickness sensitivity is conducted to examine how the throat factor of safety responds to wall thickness, providing the trade-off basis to fall back on if the baseline FoS proves marginal. The study is varied across t = 3, 4, and 5 mm at the converged Mesh 3, while maintaining the constants from CFD source data, boundary conditions, material, and mesh refinement parameters. The inner wall (gas-side) geometry is identical across all three cases with only the outer wall offset changes.

7.2 Results

The sensitivity results across three wall thicknesses are presented in the table below:

QoI	t = 3 mm	t = 4 mm	t = 5 mm
$\sigma_{vM_{throat}}$, (MPa)	3.9414	4.9171	6.1812

QoI	t = 3 mm	t = 4 mm	t = 5 mm
$\sigma_{vM_{global}}$, (singularity, MPa)	486.95	430.58	484.70
T_{throat} (K)	803.59	803.59	803.59
ΔT_{throat} (K)	1.33	2.10	2.71
δ_{exit} (mm)	1.7159	1.7150	1.7150
FoS (against σ_y = 209.95 MPa at the throat temperature of 530.44 °C)	53.27	42.70	33.96

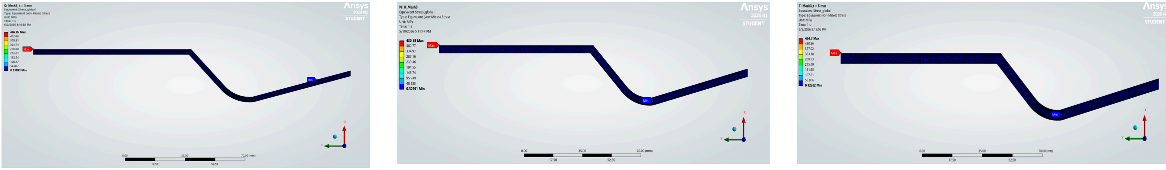
The through-wall temperature difference ΔT_{throat} rises with thickness (1.33 to 2.10 to 2.71) and tracks the conduction prediction at each thickness within 4%. It is an expected behaviour: a thicker wall presents a longer conductive path between the imposed inner-wall and the outer surface, hence a larger temperature drop develops across it.

Exit deformation δ_{exit} is essentially thickness-independent consistent with free axial expansion. Both results confirm thermal solve and structural compliance are correctly resolved across all three configurations, and serve as positive controls that the setup behaves physically.

The most important observation in these results is the reversed relationship of throat stress and wall thickness. A thin-wall pressure-vessel reading predicts throat stress should fall with thickness, since hoop stress scales with P (pressure) and r (radius), not t (thickness). The FEA shows the opposite: $\sigma_{vM_{throat}}$ rises with thickness from 3.94 to 4.92 to 6.18 MPa. The throat is therefore not pressure-hoop dominated, but rather by thermal-bending stress arising from constrained differential expansion through the wall, and because ΔT_{throat} grows with thickness, the thermal component increases faster than the pressure component falls. The net von Mises follows the growing thermal component. This inversion is the central finding of the sensitivity study.

All three configurations are far from yield-limited. The evaluation against the temperature-adjusted yield strength for all three thickness gives a FoS above 33. Wall thickness selection at this operating point is therefore not governed by yield. As established for the baseline, the binding constraint remains creep: the throat temperature is identical across all three cases and sits above SS316 creep threshold.

The global maximum von Mises stress (486.95, 430.58, 484.70 MPa) is the constraint singularity at the inlet vertex and is not a quantity of engineering interest; its non-monotonic variation across thickness reflects minor inlet-edge geometry changes and mesh-discretisation effects at the singular point, and is reported only for completeness.



t = 3 mm   t = 4 mm   t = 5 mm

Figure 21. von Mises stress contours at t = 3, 4, 5 mm

7.3 Interpretation

This section interprets the two focal points of the sensitivity study: the trend reversal is physically explicable and the artefact of the boundary condition.

Treating the throat as a thin-walled pressure vessel hoop stress scales as Pr/t and therefore decreases with thickness (5.29, 3.96, and 3.17 MPa at t = 3, 4, and 5 mm). If the throat were pressure-dominated, the thinnest wall would be the most stressed. However, the FEA shows the opposite, with $\sigma_{vM_{throat}}$ increases with thickness (3.94, 4.92, 6.18 MPa). Thus, the throat is not pressure-dominated.

The mechanism is thermal-bending stress at the throat. The throat is a geometric inflection point where the wall is unable to expand freely. A temperature difference across the wall is reacted as a bending stress at the inner surface, which is where the von Mises stress probe is located. A thicker wall presents a longer conductive path from the hot inner surface to the cooler outer surface, so the through-wall temperature difference grows with thickness. This also implies that when thickness increases, the growing thermal-bending stress outpaces the falling pressure-hoop stress, consequently increasing the new von Mises stress.

The caveat is that this trend reversal is amplified by the imposed wall-temperature BC. The thermal model fixes the inner-wall temperature at the adiabatic-wall value regardless of wall thickness, resulting in the configuration being forced to the same inner-surface temperature, and the full effect of thickness falls on the through-wall temperature difference. Alternatively, to rectify this trend, the Bartz convection boundary condition would suffice this situation as it behaves differently. It would allow the inner wall to run cooler as thickness increases since a thicker wall conducts heat away from the gas-side surface more effectively, compressing the thickness-dependence of ΔT and weakening the trend reversal.

The headline result that recovers from this caveat is the factor of safety. FoS exceeds 33 at all three thicknesses; therefore, none of the configurations is close to yield-limited. Wall thickness selection at this operating point is governed by creep life and manufacturing minimums, not by yield strength.

8. Discussion

8.1 Engineering Answer

Uncooled SS316 is not yield-limited at this operating point: FoS = 42.7 against temperature-dependent yield at the 4 mm thickness baseline. The sensitivity study confirms this conclusion is robust across t = 3, 4, and 5 mm, holding the FoS > 33. However, yield is not the binding constraint. The wall sits at 530.44 °C, which is above the SS316 creep threshold of ~410 °C. For sustained operation, time-dependent creep is the governing limit, not yield. Thus, the uncooled SS316 is viable for a short-duration prototype or ground-test firings at this operating point, but for further progress on sustained operation or flight qualification, creep analysis is required.

8.2 Physical Interpretation of Results

The factor of safety is high across all thicknesses because the throat is lightly loaded at this operating point. Two elements explain this reading: the magnitude and the dominating mechanism.

The magnitude is small because the outer-wall convective resistance dominates the heat path. With an outer wall coefficient of $10\ W/m^2 \cdot K$, the outer convective resistance exceeds the wall's conductive resistance by over two orders of magnitude. The entire inner-to-ambient temperature difference is shed at the outer surface and only ~2% appears as a gradient through the wall. Thermal stress scales with that temperature difference ΔT, and this small difference produces only a few MPa of stress. Even at the limiting wall t = 5 mm, the throat von Mises stress reaches just 6.18 MPa, far below the temperature-adjusted yield strength of $\sigma_y = 210\ MPa$. Additionally, the pressure-hoop stress is of the same small order. Neither thermal nor pressure loading is severe at the throat, hence FoS against yield is large.

The dominant mechanism is a separate question. Among these individual small stress contributions, thermal-bending is the largest, exceeding that of the pressure-hoop stress. This is identified by tracking the direction of the thickness trend relative to which stress is dominant, and the results demonstrate that thermal-bending stress increases and pressure-hoop stress decreases with increasing wall thickness, implying the ranking of these contributing stresses, not their absolute size. This dominance is specific to the imposed-temperature BC. Under Bartz convection BC, the through-wall temperature difference would shrink with increasing thickness, and pressure-hoop would likely reclaim dominance at thicker walls.

8.3 Literature Context

The finite-element structural analysis of converging-diverging nozzles is well established. Mohan [1] analysed a 2 MPa internal-pressure load on a steel C-D nozzle in ANSYS Mechanical, reporting peak von Mises at the throat and flange transition and identifying coupled thermal-structural analysis as necessary future work; the present study extends that pressure-only foundation into the combined pressure and thermal regime at the same chamber pressure. The one-way coupling used here, which maps CFD-computed pressure and temperature fields onto the structural model, follows established rocket-nozzle practice [2], and the verification of CFD throat conditions against isentropic calculations reflects a recognised CFD nozzle approach [5]. The thermal bending response that governs the present results, in which a through-wall temperature gradient produces bending stress under internal restraint, is consistent with classical thermoelastic theory for restraint walls [3]. Stainless steel has direct precedent as an uncooled nozzle material in hydrogen-peroxide monopropellant thrusters: Lo Sapio et al. [4] ground-tested a 20 N thruster whose products expand through an uncooled stainless-steel convergent-divergent nozzle at chamber temperatures approaching 900 °C, and SS316 specifically has been used as the structural material in comparable small monopropellant thrusters [9]. Neither, however, reports a structural or thermal-stress analysis of the nozzle, and both operate at lower chamber pressure than the present case. The contribution here is therefore not the uncooled stainless-steel configuration itself — which has been built and fired — but a quantified thermal-structural feasibility assessment of it at this operating point, which has no published precedent that was identified.

9. Known Limitations

#	Limitation	Impact	Mitigation / Note
L1	y+ peak ~18 at throat (buffer zone)	Reduced near-wall accuracy for k-ω SST at throat	Documented; bulk validation passes 5% gate
L2	T_aw slightly exceeds T_c at throat (~1%)	Numerical artefact — thermodynamically T_aw ≤ T_0 for r < 1	Conservative for FEA thermal load
L3	Residuals floor at ~10^{-3}	Steady RANS limit-cycle at transonic conditions	Mass flow imbalance (0.04%) is the primary criterion; passes by 50× margin
L4	Adiabatic wall CFD (one-way FSI)	No structural deformation feedback into CFD; no wall conduction in CFD	Standard for preliminary analysis; low-outer wall heat flux confirms wall conduction does not change gas-side T_aw (gas-side Bartz Bi ≈ 0.6; outer-wall convective resistance is greater than wall conductive resistance, so the gas-side T_aw is unaffected by wall conduction)
L5	Outer wall h = 10 W/m²·K (still air)	Real test stand may have additional convection from exhaust plume	Conservative; underestimates outer wall cooling; FoS is a lower bound
L6	2D axisymmetric (no 3D effects)	Ignores any non-axisymmetric instabilities	Geometry and loading are axisymmetric; also license-driven
L7	Steady-state only (no transient startup)	Startup transients may produce higher peak loads	Out of scope per §2.3; flagged as future work
L8	No creep model	Wall above SS316 creep threshold (~410°C)	Active scope boundary. FoS = 42.7 against yield is NOT the binding limit
L9	Negative gauge pressure at exit edge (~−40 kPa)	Local numerical artefact at the corner cell	Bulk area-weighted exit pressure (61.5 kPa) is correct; corner excluded from FEA

#	Limitation	Impact	Mitigation / Note
L10	Three-thickness sensitivity sweep (3, 4, 5 mm)	Trend identification, not optimisation; no resolution within range	Adequate for the engineering conclusion (yield not binding); finer sweep is future work
L11	Imported Load freshness — single-node CFD export variability	~3% on throat stress, ~3% on FoS	Caught in Phase 4 v3; freshness gate added (Min/Max baseline check at every solve)
L12	Imposed wall temperature BC (Dirichlet thermal)	Amplifies t-dependence of throat stress in the sensitivity study	Bartz convection BC follow-up identified as Future Work item 1

10. Future Work

There are five future works identified after the analysis and results achieved:

1. Convective inner-wall boundary condition
    a. Replace the imposed fixed-temperature BC with a Bartz convection model to remove the BC artefact in the thickness sensitivity study.
2. Creep assessment at the throat
    a. Results show that yield is not the binding constraint and the throat temperature of ~531 °C exceeds the SS316 creep threshold of ~410 °C. The creep assessment will determine the real service limit of the nozzle.
3. Specified-minimum material bound
    a. FoS uses typical yield values, therefore, re-running against ASME spec-minimum would give a fully conservative figure.
4. External radiation
    a. It was omitted to follow a conservative assumption (wall runs hotter without it). If included, the outer surface radiation BC would refine the outer-wall temperature and the through-wall gradient.
5. Constraint singularity at the free-end outer wall inlet
    a. Identified as an artifact as global von Mises stress diverges with mesh refinement. Revised constraint configuration or local treatment would let the global field be reported, rather than relying on the throat-probe values.

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